Finite Groups and Their Representations — Exercise Sheet 12

Tableaux and Young's orthogonal form

Variant: Questions only

Exercise 1 (Dimensions from branching). Prove $f^\lambda = \sum_{\mu=\lambda-\square} f^\mu$ for $|\lambda| \geq 2$ by removing the box containing the largest entry. Start with $f^{(1)} = 1$ and compute every dimension for S_6 without using the hook-length formula. Check that their squares sum to 6!. Recall from Lecture 12 that $f^\lambda = \dim V^\lambda$.

Exercise 2 (The five-dimensional representation of S_5). List the standard tableaux of shape $(3,2)$ in the following order: T_1 has first row 123, T_2 has 124, T_3 has 125, T_4 has 134, and T_5 has 135. Write all four generator matrices on the orthonormal basis $(v_{T_1}, \ldots, v_{T_5})$ with the positive off-diagonal convention of Lecture 12. Verify the braid relation for s_2, s_3. Compute the characters at $(1\,2)$, $(1\,2\,3)$, $(1\,2)(3\,4)$ and $(1\,2\,3\,4\,5)$.

Exercise 3 (GZ idempotents). For $n \geq 1$, put $C_k = \{-(k-1), \ldots, k-1\}$. For every standard tableau T with n boxes define

$$E_T = \prod_{k=2}^{n} \prod_{\substack{c \in C_k \\ c \neq c_k(T)}} \frac{X_k - c}{c_k(T) - c}.$$

For $n = 1$ the empty product is e. The factors commute because the JM elements commute by Lecture 10. Prove that C_k contains the spectrum of X_k on every irreducible, and that E_T acts as projection onto $\mathbb{C}v_T$, and as zero in the other irreducibles. Prove that the E_T are pairwise orthogonal primitive idempotents whose sum is e, and that $\sum_{T \in \mathrm{SYT}(\lambda)} E_T = e_{V^\lambda}$. Find the rank of left multiplication by E_T in the regular module. Compute the two idempotents at $n = 2$ in the basis $e, (1\,2)$. Here primitive means not a sum of two non-zero orthogonal idempotents.

Exercise 4 (Transposed shapes and the sign representation). For $T \in \mathrm{SYT}(\lambda)$ let T' be its transpose and put $\varepsilon(T) = \mathrm{sgn}(\sigma_T)$, with σ_T as in Lecture 12. Use Young's orthogonal bases of Lecture 12 for both shapes, with positive off-diagonal coefficients. Prove that $v_T \mapsto \varepsilon(T) v_{T'}$ intertwines V^λ with $\mathrm{sgn} \otimes V^{\lambda'}$. Deduce $V^{\lambda'} \cong \mathrm{sgn} \otimes V^\lambda$. Use Lecture 12's matrices for $(3,1)$ to give the matrices for $(2,1,1)$ in the corresponding signed-transpose basis. For the self-conjugate shape $(2,2)$, compute the signed-transpose operator in the ordered basis (v_{Q_1}, v_{Q_2}), where Q_1, Q_2 have first rows 12 and 13, respectively, and square it.

Exercise 5 (The content sum of an irreducible). Prove that $\sum_{k=1}^{n} X_k$ acts on V^λ by $\sum_{b \in \lambda} c(b)$. Use this to compute its scalars for $\lambda = (3,2), (2,1,1), (2,2)$ and the corresponding character values at a transposition. Use dimensions $5, 3, 2$, respectively. Compare these values with Exercises 2 and 4 and Lecture 12's $(2,2)$ self-check. For $n = 1$ state the scalar without referring to a transposition.

Exercise 6 (Tableau inversions and Coxeter length). The Coxeter length of $\sigma \in S_n$ is the smallest number of adjacent transpositions whose product is σ. A word with this minimum number of factors is called *reduced*. Prove that this length for σ_T equals $\mathrm{inv}(T)$

from Lecture 12. An ordinary inversion of a word $(w_1, \ldots, w_n)$ is a pair of positions $p < q$ with $w_p > w_q$. Use these inversions of the one-line word $(\sigma_T(1), \ldots, \sigma_T(n))$ for the lower bound. For the upper bound, use Lecture 12's lemma on moves and inversions: every non-row tableau has a standard move lowering $\mathrm{inv}(T)$ by one. For the tableau of shape $(3, 2)$ with first row 135, find a reduced word.

Exercise 7 (⋆ Norms of seminormal vectors). Let u_T be the seminormal vectors constructed in Lecture 12, starting with a unit vector at the row tableau. Prove the formula

$$\|u_T\|^2 = \prod_{\substack{j<k \\ k \text{ lies in a higher row than } j \text{ in } T}} \left(1 - \frac{1}{(c_k(T) - c_j(T))^2}\right).$$

Show that every denominator is non-zero and every factor positive. Prove the formula by induction on $\mathrm{inv}(T)$, using Lecture 12's lemma on moves and inversions, which gives a standard move lowering $\mathrm{inv}(T)$ for every non-row tableau, and Lecture 12's seminormal norm relation $(*)$. Compute all five squared norms for the ordering of shape $(3, 2)$ in Exercise 2, and the matrix of s_3 on (u_{T_1}, u_{T_2}).